

Data-Driven Transit Design with Network Analysis

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Transportation networks are fundamental to the functionality and growth of urban areas, shaping the efficiency of city operations and accessibility for residents. This project analyzes a network of 933 nodes and 2,950 links in the Chicago region to identify critical intersections and demonstrate how network analysis can inform strategic public transit expansion. Our methodology includes mean degree analysis, centrality metrics, network failure simulations, and origin-destination demand analysis, among others. Results reveal that each analytical approach highlights different aspects of the network—such as connectivity, vulnerability, and congestion points. Notably, incorporating traffic demand data enables us to simulate targeted public transit improvements in high-need areas. The proposed framework offers a scalable approach that can be adapted for transit planning in other metropolitan regions.

Transportation Networks | Betweenness Centrality | Urban Mobility | Shortest Path Analysis

Introduction

Public transportation is a cornerstone of urban mobility, responsible for reducing congestion, lowering emissions, and improving accessibility. Especially in rapidly growing cities, transportation is essential for driving social, economic, and cultural dynamics (1). Yet, many major U.S. urban centers lack reliable, efficient transit networks that people can depend on. The challenge involves not only expanding public transportation, but also implementing strategic design. It is important to consider the location of new transit lines in order to maximize impact. Network science offers powerful tools to tackle this problem. This project explores how network measures can guide the introduction and expansion of public transit systems in cities, specifically through the analysis of a Chicago transportation network (2). Through key metrics such as betweenness centrality and shortest path measures, we aim to identify optimal locations for new transit infrastructure (3). We deviate from some standard centrality measures, which typically operate under the assumption that only shortest paths contribute to node importance. In a traffic context, we also aim to consider the routes that minimize "cost", thus incorporating demand and congestion into our analysis. Our findings will demonstrate how network-driven insights can serve as a scalable framework for improving transit planning in cities across the globe.

Methods & Materials

Dataset Description. This study utilizes data from Brandon Stabler's Transportation Networks dataset (2), which provides a realistic yet aggregated representation of the Chicago region's road network. The dataset contains 4 files formatted in TNTP (Transportation Network Test Problems): base network data, origin demand data, node coordinates (in the Illinois State Plane coordinate system), and a best-known flow solution. For our analysis, we will primarily focus on the first three datasets.

The base network dataset defines the connectivity between nodes, with each row representing an edge that links a starting node to an ending node. In addition to connectivity, the dataset includes various attributes/weights such as capacity, length, power, toll, and other relevant features.

The origin demand dataset captures the travel demand between different zones within the network. It specifies the volume of trips originating from one zone and terminating in another, reflecting commuting patterns, traffic flow, and regional connectivity. This dataset plays a crucial role in transportation modeling, as it informs route assignment, congestion analysis, and infrastructure planning.

Finally, the node coordinate is a map that tells us where each node in the network is geographically in Illinois.

Network Representation. Chicago's road network is modeled as a directed, weighted, spatial graph $G = \{V, E\}$ of 933 nodes and 2950 edges. The edges (E) correspond to street

Significance Statement

This study uses network science to inform urban transit planning. By applying centrality and shortest path analysis to Chicago's transportation network, we propose a scalable, data-driven framework for transit optimization in growing cities.

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Shilpa: I conducted betweenness centrality and OD analyses to identify key roads and traffic demand patterns. Based on this, I simulated new metro links for the most important origin-destination pairs. Comparing the network before and after adding the links, using average shortest path length, showed an improvement. The proposed links were similar to Chicago's actual transit network.

Jordyn: I performed centrality analysis on the dataset, computing values and creating visualizations for degree and betweenness centralities. I then analyzed the hub and authority scores of the nodes and computed the top hubs and top authorities. I plotted the distributions of both of these metrics as well. I wrote the background information about hubs and centralities for the report, as well as the results of this analysis, the abstract, and our final conclusions.

Anika: I completed the base network analysis both in the project research and the write-up section. I brainstormed different metrics we could use to learn more about the structural nature of the network, such as strongly and weakly connected components, connectance, k-cores, and mean density. I then wrote code to compute these metrics, and interpreted the results of each in context of our transportation network in the write-up. I also analyzed how the mean degree changes in context of location of the city.

Gianna: I performed a resilience analysis on the Chicago network. To do this, I first investigated different methods to analyze the network and found that I could analyze the connectivity of the network using targeted and random attacks. I implemented this by writing code using the NetworkX library in python to find the betweenness of the nodes and other methods to test the network resilience. I then plotted the connectivity of the network with increasing targeted and random attacks.

Erin: I reviewed available datasets and selected one in TNTP format with coordinate data, allowing us to map our network analysis onto Chicago for greater spatial relevance. I led initial data processing and graph construction with Shilpa using NetworkX and other Python libraries, and conducted a literature review on centrality metrics to guide our analytical approach.

There are no known competing interests at the time of this report.

¹S.B.(Shilpa Bojjireddy), J.F. (Jordyn Fuchs), A.M. (Anika Misra), G.P. (Gianna Pedroza), and E.W. (Erin Wu) contributed equally to this work.

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segments, each assigned a weight that reflects factors such as travel time or capacity. The nodes (V) represent intersections or endpoints of road segments and are mapped using (x, y) coordinates from the Illinois State Plane system, ensuring geographic accuracy. Additionally, the dataset includes 387 zones, geographical areas which represent the origin and destination points for travel demand.

Base Network Analysis. For the base network analysis (4), we focused solely on the network structure, utilizing only the base network dataset. We began by making an adjacency matrix, A, defined as such:

$$A_{i,j} = \begin{cases} 1 & \text{if there exists an edge from } j \text{ to } i \\ 0 & \text{otherwise} \end{cases}$$

Because this is a directed graph, we determined the number of strongly connected and weakly connected components. A **strongly connected component** is a set of nodes where each is reachable from all the others via some directed path. A **weakly connected component** has a similar definition but does not take into account the direction of the edges (5). Both the number of strongly and weakly connected components in the network were 1. This implied that every node in the network was connected to every other node via some path both when ignoring edge directions and taking them into account, which makes sense for a transportation network: if there is a path from node A to node B, there should also be a path back from node B to node A.

With these results, we checked to see if the Adjacency matrix was symmetric, and it was. This meant that for every directed edge from a node i to a node j , there was always a corresponding directed edge in the opposite direction from node j to i . In other words, we could simply treat the network as an undirected one, with each edge representing a bi-directional connection between two road intersections. Hence, we modified the adjacency matrix A to be defined as such:

$$A_{i,j} = \begin{cases} 1 & \text{if there is a bidirectional road from } i \text{ to } j \\ 0 & \text{otherwise} \end{cases}$$

Next, we computed some key structural network properties. The **connectance** (5) of a network is defined as:

$$\rho = \frac{m}{\binom{n}{2}} \quad [1]$$

where:

- m is the number of edges in the network,
- n is the number of nodes in the network, and hence
- $\binom{n}{2}$ is the total number of possible edges in the network.

The connectance of this network, given by Eq. (1), was calculated to be approximately 0.00339. This tells us that the network structure is not very dense, which makes sense for a transportation network, as most nodes will only be connected to a few other nodes at most.

Moving onto group structure, a **k-core** of a network is defined as a set of nodes where each is joined to at least k other nodes in the set. (5). The maximum value of k for the

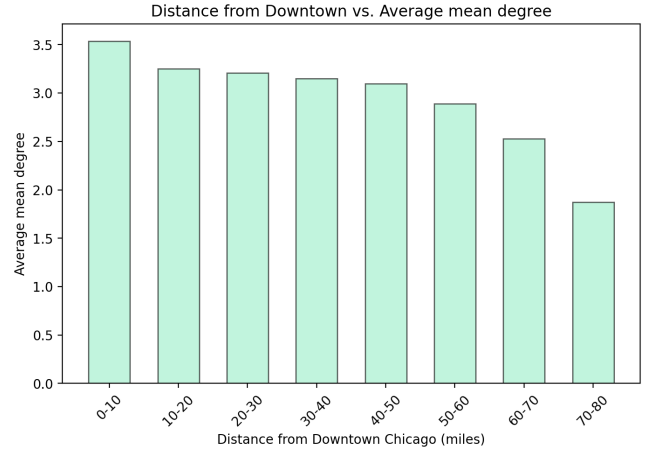


Fig. 1. How Mean Degree Changes as We Get Further from Downtown

k -core in this network was $k=3$, with 475 nodes in this 3-core. Typically, a higher k -value with a smaller number of nodes indicates that these few nodes are part of a highly cohesive group compared to the rest of the network. One could interpret this highly cohesive group as an important, highly-connected, exclusive subgroup of the graph. However, since approximately half of the nodes in the network are part of this group, and the k -value is relatively low, there is no relatively highly-connected, exclusive subgroup of intersections. Once again, these results make sense for a transportation network, as almost all of the intersections are only connected to a few others.

The average degree of the nodes in the network was 3.16, meaning that on average, each intersection was connected to approximately 3.16 roads. However, different regions of the city may have different numbers of road connections. We can see the plot of mean degree vs. node distance from downtown in Figure 1. Indeed, we see that the closer a node is to downtown, the higher its average mean degree. Nodes that were 70-80 miles from downtown had an average mean degree of approximately 2.0, which is 1.5 roads less than the nodes closest to downtown.

Network Resilience. The resilience analysis (4) of the Chicago network helps us to understand its structural robustness against node and edge failures. The analysis was performed by simulating both random and targeted failures. Random failures involved the removal of nodes or edges at random, while targeted attacks were based on the betweenness centrality of the nodes and the edges that facilitate the highest number of shortest paths. In Figure 2, which analyzes node failures, the blue line representing random failures declines more steeply than the orange line of targeted attacks. This indicates that the removal of peripheral nodes causes the network to fragment more rapidly. Meanwhile, the network is more resilient to the removal of high-centrality nodes due to alternative routes that travelers can take. In contrast, Figure 3, which examines edge failures, shows that random edge removals lead to a dramatic and sudden collapse around the eighth removal, while targeted attacks have little effect on the connectivity of the network. The sudden drop in the random edge failure graph highlights the existence of key edges that act as vital connections, while the gradual decline

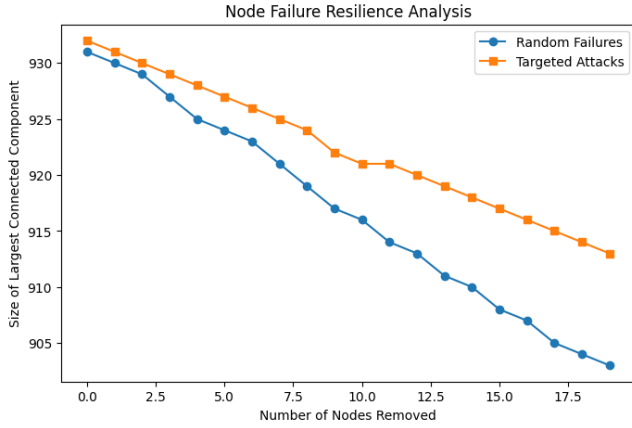


Fig. 2. Node Failure Graph

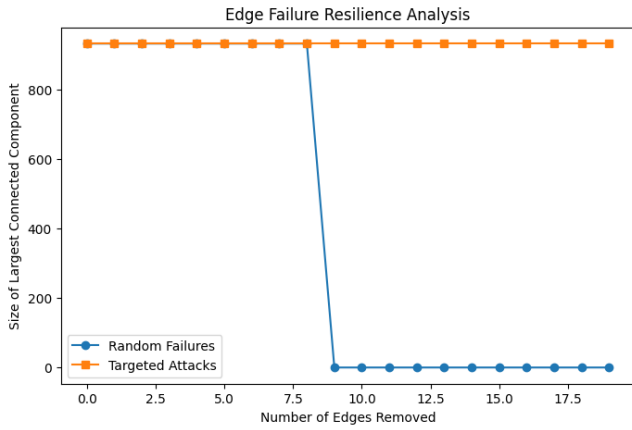


Fig. 3. Edge Failure Graph

in the node failure graph underscores the distributed nature of the network's resilience. This means that the network is heavily dependent on key intersections rather than individual roads. Even if high-capacity roads are removed, alternative routes keep the network functional, but if certain critical nodes fail, the entire transportation system can collapse. (6)

Betweenness Centrality. Centrality and shortest path metrics serve as fundamental tools in network analysis, providing insights into connectivity, efficiency, and congestion. Betweenness centrality measures how often a node appears on the shortest paths between other nodes in a network.

The **betweenness centrality** (5) of a node i is defined as:

$$x_i = \frac{1}{n^2} \sum_s \sum_t \frac{n_{s,t}^i}{g_{s,t}} \quad [2]$$

where:

- n is the total number of nodes in the network,
- $n_{s,t}^i$ is the number of times node i appears on the shortest path between nodes s and t ,
- $g_{s,t}$ is the number of geodesic (shortest) paths between s and t .

In unweighted graphs, shortest paths are computed using Breadth-First Search (BFS), while in weighted graphs, Dijkstra's algorithm is employed (5). We treat our graph as unweighted when computing betweenness centrality and weighted when analyzing shortest paths.

In our case, we extend betweenness centrality to edges rather than nodes, this will measure how often a road segment appears on the shortest paths between all node pairs. This allows us to identify key transfer roads and potential congestion points. The initial betweenness centrality computation is performed on the transportation network without incorporating traffic flow data, meaning it provides a purely structural perspective on critical roadways.

Roads with high betweenness scores are likely to be major routes within the network, as they frequently appear in shortest paths. These roads exhibit a high level of dependency, meaning that their removal would significantly disrupt connectivity between various regions. Even without explicit traffic data, intersections and roads with high betweenness scores can serve as preliminary indicators of potential congestion points or bottlenecks.

Hubs and Authorities. While some centrality measures attribute high importance to nodes that are pointed to by nodes with high centrality, it can also be useful to consider the importance of nodes that point to many other nodes with high centrality. In order to consider both of these perspectives, we define authorities as nodes that are pointed to by nodes with high hub centrality, and hubs as nodes that point to many nodes with high authority centrality.

Following the Kleinberg approach, we define the **authority centrality** (5) of a node i as proportional to the sum of the hub centralities of the nodes that point to it:

$$x_i = \alpha \sum_j A_{ij} y_j \quad [3]$$

where:

- α is a constant,
- A_{ij} is the element of the adjacency matrix corresponding to an edge from node j to node i ,
- y_j is the hub centrality of node j .

Similarly, we define the **hub centrality** (5) of a node i as proportional to the sum of the authority centralities of the nodes that it points to:

$$y_i = \beta \sum_j A_{ji} x_j \quad [4]$$

where:

- β is another constant,
- A_{ji} is the element of the adjacency matrix corresponding to an edge from node i to node j ,
- x_j is the authority centrality of node j .

Shortest Path Origin-Demand. The network analyses we've explored so far focus entirely on structure—nodes, links, and metrics like centrality—yet they ignore a crucial layer: the spatial distribution of travel demand that dictates real-world network performance. While the previous methods have highlighted roads that are important in terms of shortest path routing, they do not necessarily indicate which roads experience the highest traffic volumes in practice. Some roads with high betweenness may not actually be congested, while others with low betweenness but high travel demand might experience significant congestion.

To validate these findings, it is important to incorporate real travel demand data using the Origin-Destination (OD) matrix (2). The OD matrix provides empirical data on the actual number of trips between different areas/zones in the network. By computing shortest paths, weighted by real travel demand, we can differentiate between theoretically important roads and those that are actually congested due to demand patterns.

To systematically measure network performance, we define Average Shortest Path Length (ASPL) as our baseline metric. We define ASPL as such:

$$ASPL = \frac{1}{N(N-1)} \sum_{O \neq D} g_{O,D} \quad [5]$$

where

- N is the total number of nodes in the network
- $g_{O,D}$ represents the geodisc (shortest) path between zones O and D , computed under free-flow conditions (i.e. assuming no congestion)

ASPL, shown in Eq. (5) provides a global measure of network efficiency, capturing how easily different areas are connected. A lower ASPL indicates a more efficient network, where trips are generally shorter. However, this metric is naïve because it does not account for actual trip demand—it treats all shortest paths equally, even if some OD pairs experience significantly higher travel volumes than others.

Using the OD matrix and the weighted network, we identify the most expensive OD pairs based on travel cost. We define the cost of an OD pair as such:

$$cost = demand_{O,D} * g_{O,D} \quad [6]$$

where

- $g_{O,D}$ represents the geodisc (shortest) path between zones O and D , computed under free-flow conditions.
- Demand is taken from the OD matrix, reflecting actual trip volume between each origin-destination pair

For each OD pair, we compute the cost given by Eq. (6) and sort them from highest to lowest. The intuition behind this approach is that OD pairs with extremely high cost represent high-demand corridors that currently lack efficient connectivity. These areas are prime candidates for public transit lines or network expansion. Based on this analysis, we propose new links in the network to reduce congestion and improve efficiency.

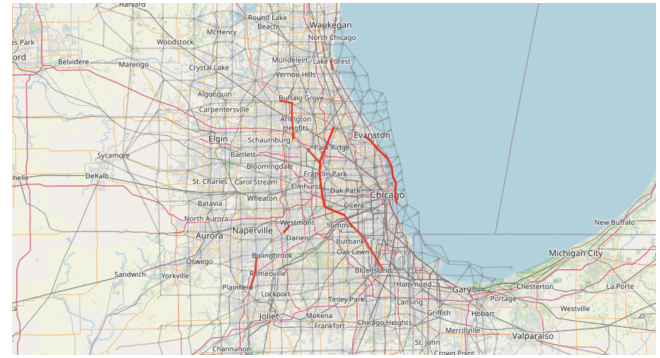


Fig. 4. Edge Betweenness Centrality of Chicago Network

Evaluating the Impact of Network Modifications. To assess whether our proposed modifications yield improvements, we recompute ASPL, given by Eq. (5) on the modified network. A reduction in ASPL indicates that the new links have successfully enhanced network connectivity, leading to shorter average trips across the system.

Results

Betweenness Centrality Results. The graph in Figure 4 illustrates the 60 highest-scoring betweenness centrality edges, overlaid onto a map of Chicago. The gray network represents all the nodes and edges in the network, while the red lines highlight the most critical edges in terms of betweenness centrality. The results indicate that downtown Chicago functions as a dominant hub, with the majority of shortest paths passing through this region. This aligns with expectations, as central business districts typically act as focal points for commuting and transportation, concentrating high volumes of travel demand. Beyond downtown, the analysis also highlights suburb-to-suburb links with high betweenness scores, revealing critical corridors that facilitate cross-regional connectivity. These are the edges oriented vertically on the map in Figure 4 traversing from northern suburbs to southern suburbs. This suggests that a few key roadways play a greater role in linking suburban areas—a pattern that may indicate structural dependencies and potential congestion risks in these corridors.

The distribution of edge betweenness scores is heavily left-skewed, with a mean of approximately 0.0030. A small number of top-ranked edges account for disproportionately high scores—some exceeding the mean by more than 15 times, including the highest-scoring edge at 0.0516. This disparity underscores that a small subset of roads disproportionately influence network flow, making them highly vulnerable bottlenecks under increasing demand. If these critical links become congested, delays will not remain localized but could propagate across the entire network, significantly impacting travel times and overall mobility.

Hub and Authority Results. The graphs in Figures 5 and 6 illustrate the distributions of the hub and authority scores for the network. We can see that the hubs and authorities have the same distributions, showing a strong right skew with a majority of the scores having low values.

Upon further analysis, we see that each of the nodes have identical hub and authority scores. Figures 5 and 6 plot

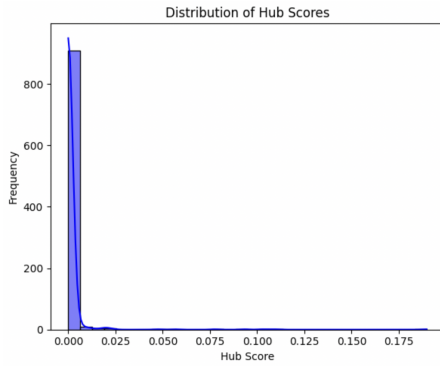


Fig. 5. Distributions of Hub and Authority Scores

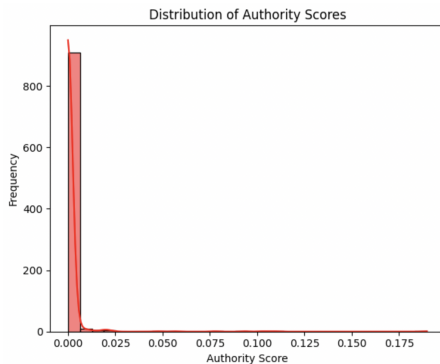


Fig. 6. Distributions of Hub and Authority Scores

these scores against their corresponding degree centralities. The important nodes act as both major hubs and major authorities. This is due to the network structure being completely symmetric, and thus essentially equivalent to an undirected network. Every directed edge has a corresponding edge in the opposite direction, so we do not gain new information when looking at hubs vs. authorities. It will be more insightful to consider the shortest paths and regions of demand in order to better understand key nodes and transit routes.

Using Origin Demand Results to add new links. Before simulating any network modifications, we first compute the Average Shortest Path Length (ASPL) given by Eq. (5) for the baseline network. The initial ASPL is 49.578, meaning that, on average, a trip between any two nodes in the network requires approximately 49.58 units of travel time under free-flow conditions. The free-flow weight reflects how long it takes a vehicle to travel along an edge assuming ideal, congestion-free traffic conditions. This provides a benchmark for evaluating network efficiency before and after link additions.

Figure 7 illustrates the worst OD pairs, highlighting the most inefficient travel routes in the current network (4). The darker edges represent OD pairs with higher cost, cost defined by Eq. (6). Notably, these worst-performing OD pairs align closely with the edges exhibiting the highest betweenness centrality. This suggests that structural properties of the network, such as shortest-path dependencies, may have a direct relationship with actual congestion patterns when

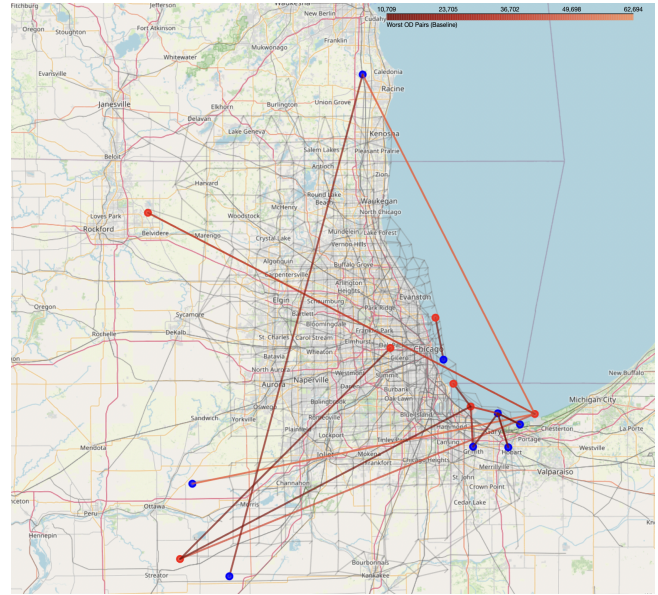


Fig. 7. Worst OD pairs overlaid on Chicago map

accounting for demand. Additionally, the node corresponding to downtown appears in 6 out of 7 of the worst OD pairs. This aligns with the intuition that downtown is a hub which experiences frequent travel demand despite having high travel costs.

Leveraging this insight, we simulate network improvements by adding links between the top six highest-cost OD pairs (4). Figure 8 shows a plot of the newly added links illustrating that they are primarily located along corridors connecting downtown Chicago to the suburbs. The edges in blue represent the added links to the network. Additionally, several of these links run perpendicular to major north-south (suburb-to-suburb) travel routes, suggesting that these additions enhance lateral connectivity across the network. These simulated links serve as a simplified representation of a metro system, providing direct, high-speed connections between high-demand OD pairs. However, this is an overly simplified approach—we assume direct travel between OD pairs without intermediate stops or congestion effects. This represents an idealized transit intervention rather than a fully detailed simulation of public transit operations.

After adding these new links, we recompute ASPL, which decreases from 49.578 to 47.50. This indicates that the network has improved, with shorter average travel distances and enhanced connectivity.

Interestingly, our OD-based link proposals closely resemble Chicago's existing metro system, see Figure 9, reinforcing the validity of using OD demand analysis as a framework for transit planning. Furthermore, our results highlight a key shortcoming of Chicago's current metro network—the lack of direct suburb-to-suburb connections. Currently, most suburban travel requires routing through downtown, increasing travel time and inefficiencies. Our findings suggest that introducing direct suburb-to-suburb transit links could significantly enhance mobility, reducing dependency on central hubs.

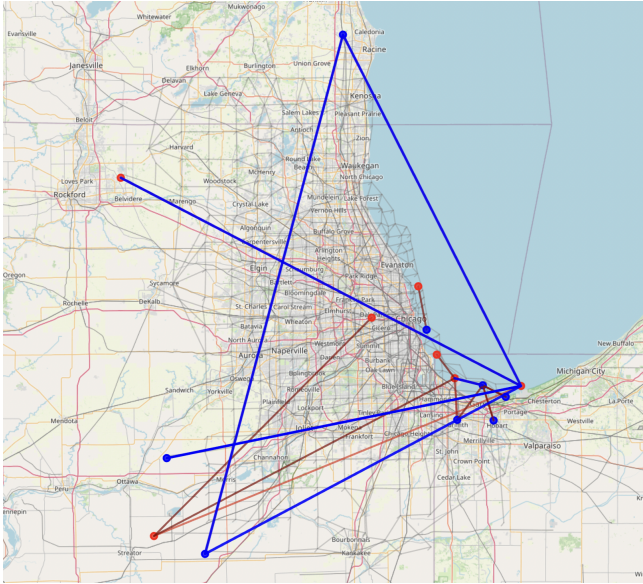


Fig. 8. Simulated Metro Links based on OD Analysis

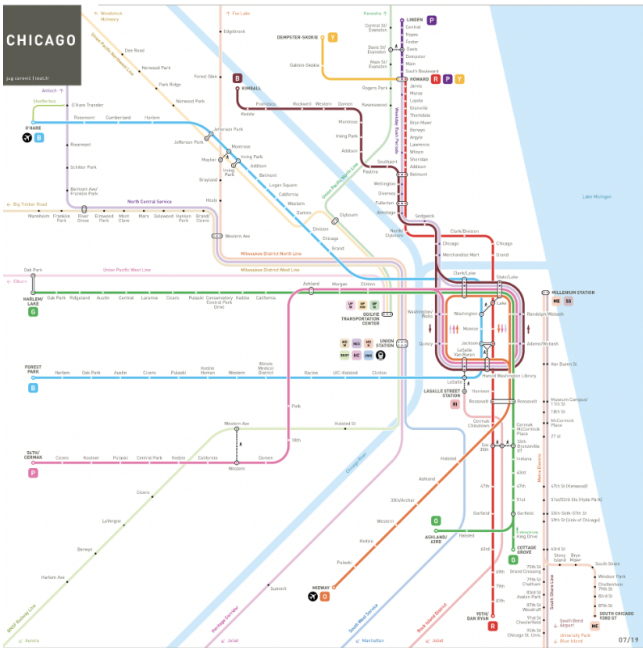


Fig. 9. Actual Chicago Metro Network

Conclusions and Discussion

The goal of our study was to develop a framework that utilizes centrality, shortest path, and demand analysis in order to improve transit planning in the Chicago region. We aimed to identify the best measures for expanding and optimizing transit networks, so that our process could be applied to other major cities. We accomplished this by researching various importance measures including betweenness centrality, hubs/authorities, and shortest path origin-demand. After applying these methods to our Chicago dataset, we were able to compute the baseline average shortest path length, identify system weak points by analyzing the worst origin demand values, and add new links to the highest cost pairs. Adding these new links helped decrease the average shortest path in the network, showing an improvement in overall connectivity.

We experienced some limitations in how complex we were able to make our analyses. Much of our cost analysis relies on simplifying assumptions, like direct transit between nodes without intermediate stops or traffic delays. Real-world traffic analysis cannot rely on these assumptions and will have to deal with dynamic congestion effects. This is something we would like to consider in future expansions on our framework. We could incorporate additional data that provides more detailed information about route demand, or consider variations in traffic depending on time of day. We would also like to apply this strategy to other major cities to see if our method will provide similar results in different regions.

Code repository

All code can be found at the following repository: <https://github.com/shilpa-bo/ChicagoTransportation>

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